

Compact model training for insurance tariff ranking

Residential building-insurance tariff ranking with SFT, a measured GRPO continuation, and production economics

Assignment outcome — achieved
The goal was to exceed 80% Top-1 agreement while producing a deployable ranked Top-3. The selected 2B SFT adapter reached 84% Top-1, 86.7% Top-3 recall, and 100% strict JSON on the held-out Fable benchmark.

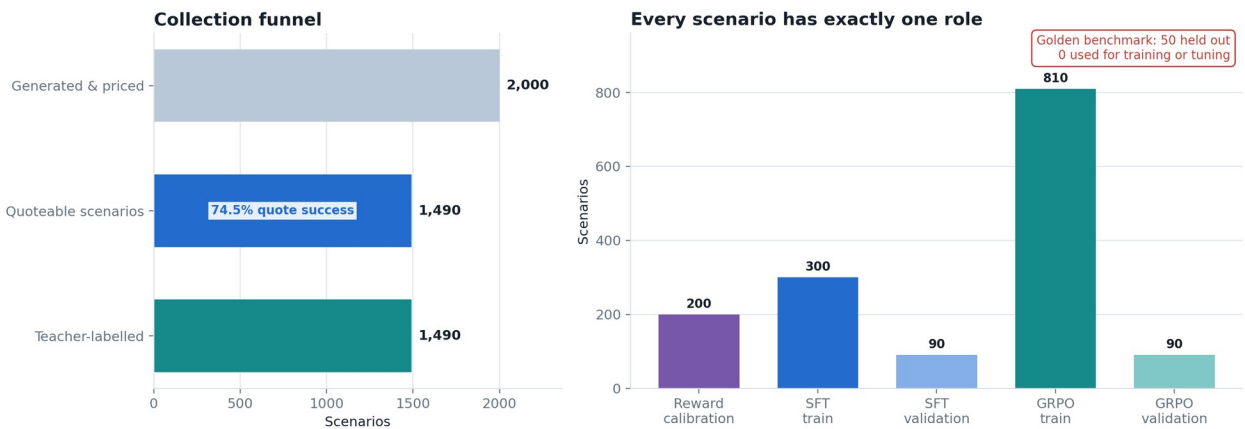
0.22 → 0.84 2B base → SFT Top-1	86.7% Top-3 recall	100% strict JSON	\$7.80 total billed spend
---	------------------------------	----------------------------	-------------------------------------

DECISION IN ONE PAGE

Task, approach, outcome

Stage	What we did	Why	Verified result
Data	Generated risks + 2,000 engine requests	Create task-specific panels	1,490 usable panels / 9,632 tariff offers
Teacher	GPT-6 Sol silver labels	Scale ranked supervision	1,490 Top-3 labels
Reward	20k deterministic search	Align proxy to teacher	94% vs GPT-6 Sol (nested CV); 84% vs Fable golden
SFT	2B BF16 LoRA	Test smaller-model hypothesis	22% → 84% golden Top-1
GRPO	10-step continuation	Test reward optimization	84% Top-1; no broad gain
Serving	RTX 4090 vLLM benchmark	Measure deployability	63.66 rec/s at c64

A deliberately separated data pipeline



Source: frozen quoted-data, split, and teacher manifests. Split counts are disjoint and exhaustive (n=1,490).

Leakage-safe path from generated risks to training and evaluation

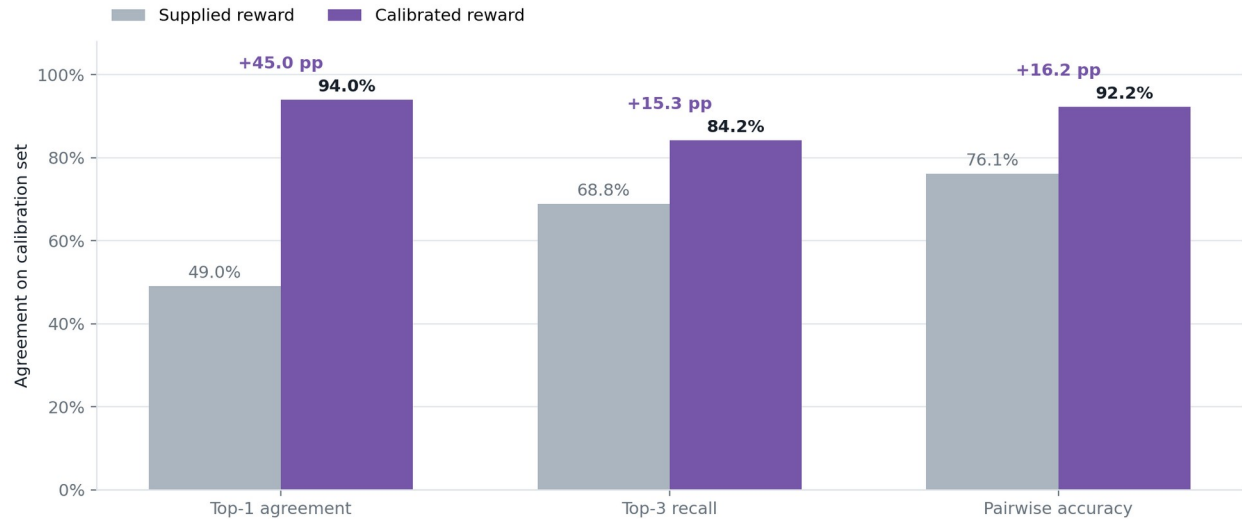
Context: the supplied Qwen3.5-9B reference scored 42% Top-1. The trained-model improvement reported here is correctly measured from our unchanged 2B base (22%) to our 2B SFT model (84%).

02 / TASK / WHAT / WHY / RESULT

Data and reward: make the objective trustworthy

Task	What we did	Why this way	What we got
Build supervision	2,000 quote-engine requests; GPT-6 Sol labelled the 1,490 usable panels	Real quote panels preserve product trade-offs; silver labels scale cheaply	9,632 tariff offers retained in usable panels; five insurers; nine tariffs; disjoint splits; zero exact golden-risk duplicates

Reward calibration makes the scorer look more like the teacher



Source: 200-scenario reward-calibration split only; 5-fold selection, no golden or GRPO-validation data. Not directly comparable to golden metrics.

Symbolic scorer agreement with GPT-6 Sol — nested 5-fold validation on 200 calibration scenarios; not Fable/golden

What the reward audit changed

Within panels, coverage count and deductible did not vary; coverage score and annual price carried the consistent ranking signal. A deterministic 20,000-candidate search selected a quality–price curve without LLM calls or golden labels. In nested five-fold validation, the symbolic scorer matched GPT-6 Sol on 188/200 calibration Top-1 choices (94%). After the weights were frozen, the same scorer matched Fable on 42/50 Top-1 choices (84%) on the separate golden set. Neither number is model accuracy.

Frozen symbolic score

$$\text{reward} = (0.2863 + 0.4273 \times \text{normalized coverage score}) / (1 + 1.7936 \times \text{normalized price})$$

Method on golden	Top-1	Top-3 hits	Top-3 recall	Inference
Calibrated symbolic argmax	42/50 (84%)	119/150	79.3%	None
Selected 2B SFT	42/50 (84%)	130/150	86.7%	Required

Why not deploy only the zero-cost scorer?

For a Top-1-only product, it is a compelling simpler baseline. The assignment requires a ranked Top-3: SFT recovered 11 more of the 150 Fable-selected positions and offers a learned policy for interactions beyond the fixed score. The tie is evidence for a strong baseline—not proof that the scorer is ground truth.

03 / LEARNED POLICY

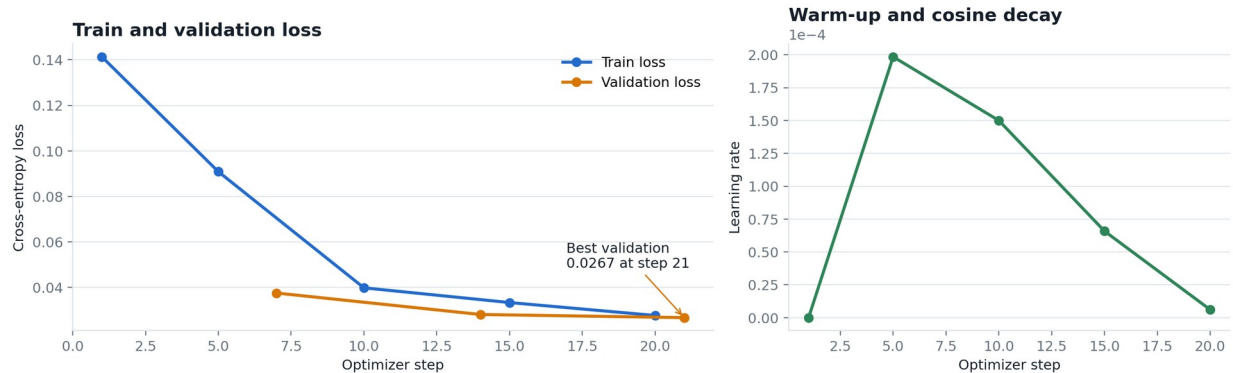
Model choice, SFT, and the GRPO stopping decision

Why 2B and why BF16 LoRA?

The proposal referenced a 9B model; our hypothesis was that structured tariff ranking should work with substantially less capacity. Qwen3.5-2B tests that claim directly. BF16 LoRA updated 10.9M parameters (0.49%): simple to run, easy to inspect, and sufficient for the available GPU. Keeping the experiment simple was also a reason not to add QLoRA complexity.

Golden evaluation	Top-1	Top-3 recall	Strict JSON	Interpretation
2B base	22%	66.7%	2%	Underpowered without task adaptation
2B + SFT	84%	86.7%	100%	Selected model
2B + SFT + GRPO-10	84%	86.0%	100%	Stable continuation; no broad gain yet

SFT converges quickly on a small, clean target



Source: frozen SFT run record, 300 train / 90 validation examples; 3 epochs; 6.3 min elapsed; 12.6 GB peak GPU memory.

SFT produced the decisive quality and format-control improvement

Why stop GRPO at ten steps?

GRPO worked technically: it remained stable and corrected one SFT Top-1 error on the 90-scenario development set (83/90 → 84/90). Ten full Top-3 orderings changed, with a small net recall loss. On golden, SFT and GRPO both reached 42/50 Top-1; GRPO changed one list but not its first choice.

Practical ceiling, not a failed method

GPT-6 Sol—the silver teacher and reward-calibration target—also achieved 84% Top-1 against Fable, and SFT had already reached that observed level while clearing the assignment’s >80% target. More GRPO steps or samples might still improve the remaining predictions, but optimizing the same silver proxy more aggressively was not justified within the time budget. SFT therefore remains the evidence-backed selection.

04 / DECISION

Production case, delivery, and conclusion

Frontier-level quality at a measured open-model cost



*2B range is measured on one \$0.74/h RTX 4090: €0.0028/1k at sustained concurrency 64 to €0.0539/1k without batching. Horizontal axis is linear. GPT-6 Sol converted from \$3.2735 to €2.8707/1k at \$1.1403/€. Reward-argmax excluded because it is not a serving model.

Held-out quality versus measured or supplied serving cost

Serving mode / comparator	Top-1	Cost per 1k	What it means
2B SFT, RTX 4090 saturated	84%	€0.0028	Measured 63.66 rec/s at concurrency 64
2B SFT, 10% utilization	84%	€0.0283	Conservative utilization scenario
2B SFT, unbatched	84%	€0.0539	Measured lower-throughput bound
Supplied Qwen3.5-9B	42%	€0.1869	Reference
Gemini / GPT-5.4	80% / 84%	€0.8897 / €1.3445	About 5–7× the 9B reference cost

Actual experiment spend

Teacher selection and batch labelling cost approximately \$3.80; the complete RunPod bill for training, evaluation, and benchmarking was \$4.00. Total billed cash spend: approximately \$7.80.

Delivered and reproducible

Git repository: <https://github.com/lakshaychhabra/reranking-optimisation>

Hugging Face dataset: <https://huggingface.co/datasets/lakshaychhabra/wg-recommendation-data>

The package includes adapters, configs, logs, frozen results, hashes, portable reproduction instructions, and prepared non-golden data. Apple MPS reproduced all 50 selected SFT outputs exactly.

Final decision

Select the Qwen3.5-2B SFT LoRA. It met the assignment objective—84% Top-1 versus the >80% target—while improving Top-3 recall, enforcing the output contract, and supporting low-cost self-hosted serving. Keep the calibrated scorer as the zero-inference Top-1 fallback and GRPO as a credible next experiment once stronger labels or a new untouched benchmark are available.

Cost scope: GPU-only serving estimates exclude storage, networking, cold starts, redundancy, and operations. The 50-scenario benchmark is held out but small; results should be confirmed on a larger expert-labelled set before production policy decisions.